



Tutor Jolly School

Name: _____

Grade: _____

Class: _____

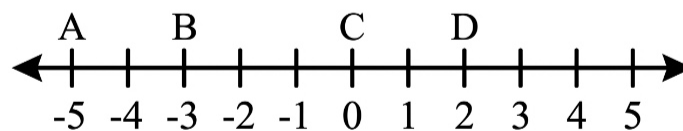
Class No: _____

NUMBER SENSE, INTEGERS, FRACTIONS, AND PRIME FACTORIZATION

Section A: Multiple Choice Questions

1. Refer to the number line below. Which letter represents the integer -3?

[2 Pts]



A. Point A

B. Point B

C. Point C

D. Point D

2. What is the prime factorization of 60?

[2 Pts]

A. $2 \times 3 \times 10$

B. $2^2 \times 3 \times 5$

C. $2 \times 3^2 \times 5$

D. $4 \times 3 \times 5$

3. Which of the following fractions is equivalent to $16/24$?

[2 Pts]

A. $2/3$

B. $3/4$

C. $4/5$

D. $1/2$

4. Round the number 14.567 to two decimal places.

[2 Pts]

A. 56

B. 57

C. 60

D. 50

5. Calculate: $-8 - (-5)$

[2 Pts]

A. -13

B. -3

C. 3

D. 13

6. What is the Least Common Multiple (LCM) of 6 and 8?

[2 Pts]

A. 12

B. 24

C. 36

D. 48

7. Evaluate: $\frac{1}{3} + \frac{1}{6}$

[2 Pts]

A. $\frac{2}{9}$

B. $\frac{1}{2}$

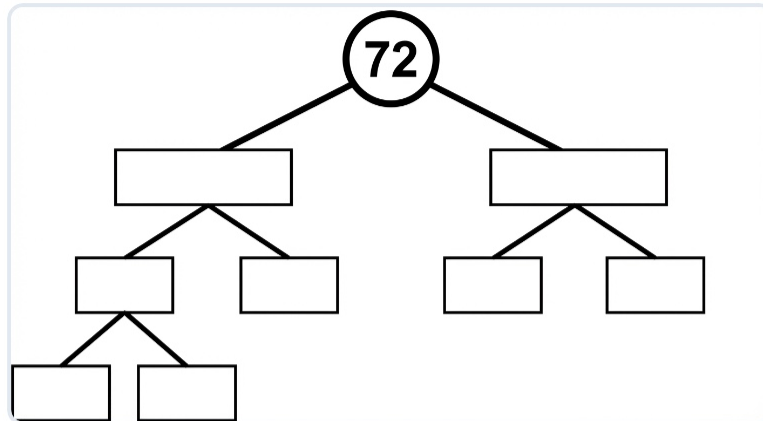
C. $\frac{2}{3}$

D. $\frac{1}{18}$

Section B: Extended Response

8. Using the factor tree method, find the prime factorization of 72. Write your final answer in index notation.

[3 Pts]

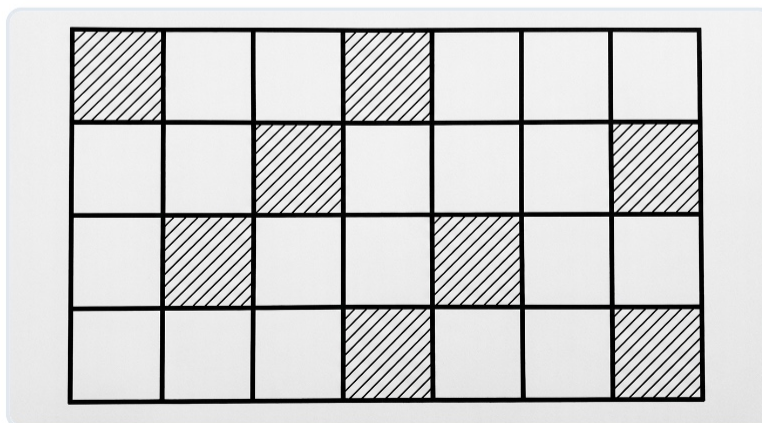


9. A florist has 36 red roses and 24 white roses. She wants to make identical bouquets with the same number of red roses and the same number of white roses in each, using all the flowers. What is the greatest number of bouquets she can make?

[3 Pts]

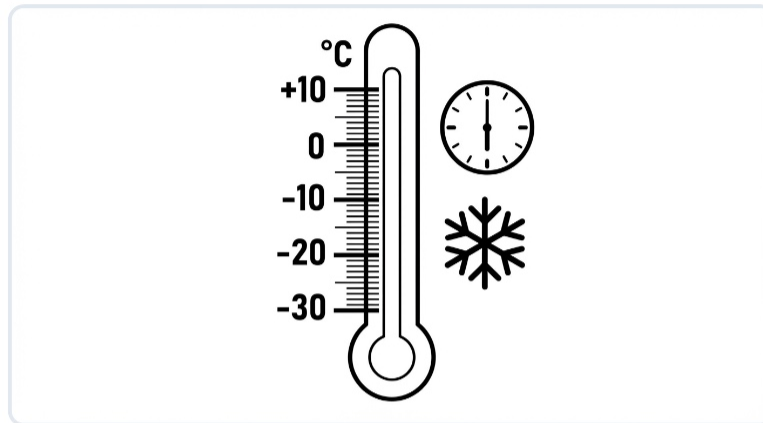
10. Calculate the total area of the shaded regions if the entire rectangle represents 1 whole unit. Give your answer as a simplified fraction.

[3 Pts]



- 11.** At 6:00 AM, the temperature in a freezer was -15°C . By noon, the temperature had risen by 4°C . By 6:00 PM, it had dropped by 7°C from the noon temperature. What was the temperature at 6:00 PM? Show your working using a number line or equation.

[4Pts]



- 12.** Evaluate the following expression, showing all steps of working:
 $12 \div (-3) + 4 \times (2 - 7)$

[4Pts]

- 13.** John spent $\frac{1}{4}$ of his money on a book and $\frac{1}{3}$ of the remainder on a toy. If he had \$20 left, how much money did he have at the beginning?

[4Pts]